A Detailed Davies-Perturbation Proof for Stationary Langevin Increments

Qian Qin

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Abstract

Let $(X_t)_{t\geq 0}$ be stationary overdamped Langevin diffusion with invariant distribution $\pi(dx) \propto e^{-U(x)}dx$ and noise coefficient $\sqrt{2}$. This note proves, for every $v \in \mathbb{R}^d$, $\lambda \in \mathbb{R}$, and $t \geq 0$,

$$\mathbb{E}\exp\{\lambda\langle v, X_t - X_0\rangle\} \leq \exp\{\lambda^2 t\|v\|^2\}.$$

The result is a standard corollary of the Davies exponential-perturbation estimate for symmetric semigroups. The purpose of the note is to supply a complete proof in the weighted Euclidean setting used by the companion MALA paper and to explain the functional-analytic terms that a brief citation normally suppresses: strongly continuous semigroup, generator domain, core, semigroup equation, conjugated generator, and closed Dirichlet form. A second part explains carefully when a deterministic-time process identity may be evaluated at a stopping time, and applies those principles to the localized stochastic-exponential calculation used in the MALA rejection proof.

Contents

1 The result and its place in the literature

Let $U : \mathbb{R}^d \to \mathbb{R}$ be continuously differentiable, assume that ∇U is globally Lipschitz, and suppose

$$Z = \int_{\mathbb{R}^d} e^{-U(x)}\, dx < \infty.$$

Define

$$\pi(dx) = Z^{-1} e^{-U(x)}\, dx.$$

On a filtered probability space carrying a d-dimensional Brownian motion B, consider

$$dX_t = -\nabla U(X_t)\, dt + \sqrt{2}\, dB_t, \qquad X_0 \sim \pi. \tag{1}$$

Global Lipschitz continuity gives a unique nonexplosive strong solution; see Karatzas and Shreve (1991, Chapter 5). The process is stationary and reversible with respect to π. One standard construction uses the closed symmetric Dirichlet form

$$\mathcal{E}(f, g) = \int_{\mathbb{R}^d} \langle \nabla f(x), \nabla g(x) \rangle \, \pi(dx), \tag{2}$$

and identifies its diffusion with the solution of (1); see Fukushima et al. (2011, Chapters 1 and 7) and Bakry et al. (2014).

The estimate needed in the MALA paper is the following.

Theorem 1.1 (Stationary linear-increment bound)**.** *For every $v \in \mathbb{R}^d$, $\lambda \in \mathbb{R}$, and $t \geq 0$,*

$$\mathbb{E} \exp\{\lambda \langle v, X_t - X_0 \rangle\} \leq \exp\{\lambda^2 t \, \|v\|^2\}. \tag{3}$$

The exact display (3) is not usually isolated as a named theorem. It is an immediate specialization of the exponential perturbation, or Davies perturbation, method for symmetric semigroups. Davies develops this method as part of the proof of Gaussian heat-kernel bounds; see Davies (1989, Chapter 3). A systematic treatment through closed forms and semigroups is given by Ouhabaz (2005, Sections 1.3–1.5 and 6.3). The proof below records the specialization with all constants and domains made explicit.

There is also a probabilistic proof. The Lyons–Zheng decomposition writes a stationary increment of a symmetric Markov process as one half of a forward martingale minus one half of a reverse-time martingale. Applied to the coordinate function $x \mapsto \langle v, x \rangle$, it gives (3) by Cauchy–Schwarz and Gaussian moment-generating functions. See Fukushima et al. (2011, Section 5.7, especially Theorem 5.7.1), the original paper of Lyons and Zheng (1988), and the later formulation of Kuwae and Uemura (1997). That route is developed in the first continuous-time companion note (Qin, 2026). The present note concentrates on the shorter analytic route.

2 A dictionary for semigroups and generators

This section gives the minimum functional analysis needed for the proof. All Hilbert spaces below are real; this avoids irrelevant complex conjugations.

2.1 Strongly continuous semigroups

Let H be a Hilbert space. A family of bounded linear operators $(T_t)_{t\geq 0}$ on H is a *strongly continuous semigroup*, or a C_0-semigroup, if

$$T_0 = I, \qquad T_{s+t} = T_s T_t, \qquad \lim_{t\downarrow 0} \|T_t f - f\|_H = 0 \quad \text{for every } f \in H.$$

The last condition is continuity in the Hilbert-space norm, not pointwise continuity in the state variable.

For a Markov process with invariant law π, the transition operators

$$S_t f(x) = \mathbb{E}[f(X_t) \mid X_0 = x]$$

form a contraction semigroup on $L^2(\pi)$. In the reversible case they are self-adjoint:

$$\langle f, S_t g\rangle_{L^2(\pi)} = \langle S_t f, g\rangle_{L^2(\pi)}.$$

For the Langevin diffusion (1), strong continuity and self-adjointness follow from the standard Dirichlet-form construction cited above.

2.2 The generator and its domain

The generator A of a C_0-semigroup (T_t) is defined by

$$Af = \lim_{h\downarrow 0} \frac{T_h f - f}{h},$$

but this limit need not exist for every $f \in H$. Its domain is

$$\mathsf{D}(A) = \left\{ f \in H : \lim_{h\downarrow 0} \frac{T_h f - f}{h} \text{ exists in } H \right\}. \tag{4}$$

The generator is generally unbounded, but it is closed and its domain is dense in H.

The basic semigroup differentiation theorem is the following. It is a standard result in any text on C_0-semigroups; see Pazy (1983, Chapter 1) or Engel and Nagel (2000, Chapter II).

Proposition 2.1 (Semigroup equation)**.** *If $f \in \mathsf{D}(A)$, then $T_t f \in \mathsf{D}(A)$ for every $t \geq 0$, and the map $t \mapsto T_t f$ is differentiable in H, with*

$$\frac{d}{dt} T_t f = A T_t f = T_t A f. \tag{5}$$

Thus the "semigroup equation" is the infinite-dimensional analogue of

$$\frac{d}{dt} e^{tA} f = A e^{tA} f$$

for a matrix A. The domain condition is what makes the derivative legitimate.

2.3 What a core is, and why the proof below does not need one

A subspace $\mathcal{C} \subseteq \mathsf{D}(A)$ is a *core* for A if it is dense in $\mathsf{D}(A)$ for the graph norm

$$\|f\|_A := \|f\|_H + \|Af\|_H .$$

Equivalently, the closure of the restricted operator $A|_{\mathcal{C}}$ is the full generator A. A core is useful when one first identifies a differential expression on a convenient class, such as $C_c^\infty(\mathbb{R}^d)$, and then wishes to conclude that its closure is the actual generator.

The proof in Section 4 does not say "take a function in a core." Instead, it works directly with the full generator domain $\mathsf{D}(A_\psi)$ of the conjugated semigroup. The semigroup equation is then available from theorem 2.1. The only differential calculation is performed through the already closed Dirichlet form, where bounded Lipschitz multiplication is legitimate.

3 The Langevin Dirichlet form

Let $H = L^2(\pi)$. The closure of (2), initially defined on $C_c^\infty(\mathbb{R}^d)$, has domain

$$\mathsf{D}(\mathcal{E}) = \left\{ f \in L^2(\pi) : f \text{ has a weak gradient and } \int \|\nabla f\|^2 \, d\pi < \infty \right\}. \tag{6}$$

This is the weighted Sobolev space $H^1(\pi)$. The associated self-adjoint generator is denoted by $\mathcal{L}$. Our sign convention is that $\mathcal{L}$ is nonpositive. For smooth compactly supported functions,

$$\mathcal{L}f = \Delta f - \langle \nabla U, \nabla f \rangle .$$

The generator and form are related by

$$\langle g, \mathcal{L}f \rangle_{L^2(\pi)} = -\mathcal{E}(g, f), \qquad f \in \mathsf{D}(\mathcal{L}), \quad g \in \mathsf{D}(\mathcal{E}). \tag{7}$$

In particular, $\mathsf{D}(\mathcal{L}) \subseteq \mathsf{D}(\mathcal{E})$.

We will repeatedly multiply form-domain functions by bounded Lipschitz functions. The needed product rule is standard for weak derivatives, but we record it explicitly.

Lemma 3.1 (Bounded Lipschitz multipliers)**.** *Let $a : \mathbb{R}^d \to \mathbb{R}$ be bounded and belong to $W^{1,\infty}(\mathbb{R}^d)$. If $f \in \mathsf{D}(\mathcal{E})$, then $af \in \mathsf{D}(\mathcal{E})$ and*

$$\nabla(af) = a\nabla f + f\nabla a \tag{8}$$

in the weak sense. In particular, if $\psi \in W^{1,\infty}$ is bounded, then multiplication by e^ψ and by $e^{-\psi}$ maps $\mathsf{D}(\mathcal{E})$ into itself.

Proof. Choose smooth functions f_n converging to f in the norm

$$\|f\|^2_{\mathsf{D}(\mathcal{E})} = \|f\|^2_{L^2(\pi)} + \mathcal{E}(f, f).$$

The weak product rule for a Sobolev function and a bounded Lipschitz function gives (8); alternatively, approximate a by smooth Lipschitz functions with the same uniform bounds. The estimates

$$\|af\|_{L^2(\pi)} \leq \|a\|_\infty \|f\|_{L^2(\pi)} ,$$

and

$$\|\nabla(af)\|_{L^2(\pi)} \leq \|a\|_\infty \|\nabla f\|_{L^2(\pi)} + \|\nabla a\|_\infty \|f\|_{L^2(\pi)}$$

show that $af \in \mathsf{D}(\mathcal{E})$. Taking $a = e^{\pm\psi}$ gives the last assertion because ψ is bounded and Lipschitz. Standard Sobolev multiplier facts of this type are discussed, for example, in Ouhabaz (2005, Chapter 1). $\square$

4 The Davies perturbation estimate

Let $\psi \in W^{1,\infty}(\mathbb{R}^d)$ be bounded. Define multiplication operators

$$M_\psi f = e^{\psi} f, \qquad M_{-\psi} f = e^{-\psi} f.$$

They are bounded inverses on $L^2(\pi)$. Define the conjugated operators

$$T_t^{\psi} = M_{-\psi} S_t M_\psi, \qquad T_t^{\psi} f = e^{-\psi} S_t(e^{\psi} f). \tag{9}$$

Lemma 4.1 (The conjugated generator)**.** *The family $(T_t^{\psi})_{t \geq 0}$ is a strongly continuous semigroup. Its generator $\mathcal{L}_\psi$ has domain*

$$\mathsf{D}(\mathcal{L}_\psi) = M_{-\psi}\mathsf{D}(\mathcal{L}) = \{f : e^{\psi} f \in \mathsf{D}(\mathcal{L})\}, \tag{10}$$

and

$$\mathcal{L}_\psi f = e^{-\psi}\mathcal{L}(e^{\psi} f). \tag{11}$$

Proof. The semigroup property follows by cancellation of the middle multiplication operators:

$$T_s^{\psi} T_t^{\psi} = M_{-\psi} S_s M_\psi M_{-\psi} S_t M_\psi = M_{-\psi} S_{s+t} M_\psi = T_{s+t}^{\psi}.$$

Strong continuity follows from boundedness of the multiplication operators:

$$\left\| T_t^{\psi} f - f \right\|_2 \leq \left\| e^{-\psi} \right\|_\infty \left\| S_t(e^{\psi} f) - e^{\psi} f \right\|_2 \longrightarrow 0.$$

Finally,

$$\frac{T_t^{\psi} f - f}{t} = e^{-\psi} \frac{S_t(e^{\psi} f) - e^{\psi} f}{t}.$$

Because multiplication by $e^{-\psi}$ is a bounded invertible operator, the limit on the left exists in $L^2(\pi)$ if and only if the limit inside on the right exists. By the definition of the generator domain, this is equivalent to $e^{\psi} f \in \mathsf{D}(\mathcal{L})$, and the limit is (11). $\square$

Lemma 4.2 (Numerical-range estimate)**.** *For every $f \in \mathsf{D}(\mathcal{L}_\psi)$,*

$$\langle f, \mathcal{L}_\psi f \rangle_{L^2(\pi)} \leq \|\nabla \psi\|_\infty^2 \, \|f\|_{L^2(\pi)}^2 . \tag{12}$$

Proof. Let $f \in \mathsf{D}(\mathcal{L}_\psi)$ and set $g = e^{\psi} f$. Then $g \in \mathsf{D}(\mathcal{L}) \subseteq \mathsf{D}(\mathcal{E})$. By theorem 3.1, both $f = e^{-\psi} g$ and $e^{-\psi} f = e^{-2\psi} g$ belong to $\mathsf{D}(\mathcal{E})$. Therefore the form-generator identity (7) applies:

$$\begin{aligned} \langle f, \mathcal{L}_\psi f \rangle_{L^2(\pi)} &= \left\langle e^{-\psi} f, \mathcal{L}(e^{\psi} f) \right\rangle_{L^2(\pi)} \\ &= -\mathcal{E}(e^{-\psi} f, e^{\psi} f). \end{aligned}$$

Using the weak product rule,

$$\nabla(e^{-\psi} f) = e^{-\psi}(\nabla f - f \nabla \psi), \qquad \nabla(e^{\psi} f) = e^{\psi}(\nabla f + f \nabla \psi).$$

The cross terms cancel pointwise, so

$$\begin{aligned} \mathcal{E}(e^{-\psi} f, e^{\psi} f) &= \int \langle \nabla f - f \nabla \psi, \nabla f + f \nabla \psi \rangle \, d\pi \\ &= \int \|\nabla f\|^2 \, d\pi - \int f^2 \|\nabla \psi\|^2 \, d\pi. \end{aligned}$$

Consequently,

$$\langle f, \mathcal{L}_\psi f \rangle = -\int \|\nabla f\|^2 \, d\pi + \int f^2 \|\nabla \psi\|^2 \, d\pi \leq \|\nabla \psi\|_\infty^2 \, \|f\|_2^2 ,$$

which is (12). $\square$

Theorem 4.3 (Davies perturbation estimate)**.** *For every bounded $\psi \in W^{1,\infty}(\mathbb{R}^d)$ and every $t \geq 0$,*

$$\left\| e^{-\psi} S_t(e^{\psi}\,\cdot) \right\|_{L^2(\pi)\to L^2(\pi)} \leq \exp\{t\,\|\nabla\psi\|_\infty^2\}. \tag{13}$$

Proof. First take $f \in \mathsf{D}(\mathcal{L}_\psi)$ and put $u(t) = T_t^\psi f$. By theorem 2.1, $u(t) \in \mathsf{D}(\mathcal{L}_\psi)$ and

$$u'(t) = \mathcal{L}_\psi u(t)$$

in $L^2(\pi)$. The Hilbert-space chain rule therefore gives

$$\begin{aligned} \frac{d}{dt}\|u(t)\|_2^2 &= 2\,\langle u(t), u'(t)\rangle_2 \\ &= 2\,\langle u(t), \mathcal{L}_\psi u(t)\rangle_2 \\ &\leq 2\,\|\nabla\psi\|_\infty^2\,\|u(t)\|_2^2\,, \end{aligned}$$

where the last line is theorem 4.2. Gronwall's inequality gives

$$\left\|T_t^\psi f\right\|_2 \leq e^{t\|\nabla\psi\|_\infty^2}\,\|f\|_2\,, \qquad f \in \mathsf{D}(\mathcal{L}_\psi). \tag{14}$$

The domain $\mathsf{D}(\mathcal{L})$ is dense in $L^2(\pi)$, and multiplication by $e^{-\psi}$ is bounded and invertible. Hence $\mathsf{D}(\mathcal{L}_\psi) = e^{-\psi}\mathsf{D}(\mathcal{L})$ is also dense. For an arbitrary $f \in L^2(\pi)$, choose $f_n \in \mathsf{D}(\mathcal{L}_\psi)$ with $f_n \to f$ in $L^2(\pi)$. Since T_t^ψ is bounded for each fixed t, $T_t^\psi f_n \to T_t^\psi f$. Passing to the limit in (14) proves (13). ☐

Remark 4.4 (Where the usual core argument went)**.** *A shorter presentation often verifies theorem 4.2 first on C_c^∞, says that this class is a core, differentiates $T_t^\psi f$, and then extends by closure. That is valid once the core statement and the passage to the conjugated generator are justified. The proof above avoids that compressed language: the Dirichlet form supplies the identity on the full generator domain, and theorem 4.1 identifies the conjugated domain directly.*

5 Proof of the stationary increment bound

We now apply theorem 4.3. Choose smooth functions $\chi_R : \mathbb{R} \to \mathbb{R}$ such that

$$\chi_R(s) = s \quad \text{when } |s| \leq R, \qquad |\chi_R'(s)| \leq 1, \qquad \chi_R \text{ is bounded.}$$

For example, one may begin with a smooth function whose derivative is one on $[-1,1]$, vanishes outside $[-2,2]$, and lies between zero and one, and then rescale. Set

$$\psi_R(x) = \lambda\chi_R(\langle v, x\rangle).$$

Then ψ_R is bounded and

$$\|\nabla\psi_R\|_\infty \leq |\lambda|\,\|v\|\,.$$

Stationarity, the Markov property, and the definition of the semigroup give

$$\begin{aligned} \mathbb{E}e^{\psi_R(X_t)-\psi_R(X_0)} &= \int e^{-\psi_R(x)} S_t(e^{\psi_R})(x)\,\pi(dx) \\ &= \left\langle 1, T_t^{\psi_R} 1\right\rangle_{L^2(\pi)}. \end{aligned} \tag{15}$$

Because π is a probability measure, $\|1\|_{L^2(\pi)} = 1$. Cauchy–Schwarz and theorem 4.3 imply

$$\begin{aligned}\mathbb{E}e^{\psi_R(X_t)-\psi_R(X_0)} &\le \left\|T_t^{\psi_R}\right\|_{2\to 2}\\ &\le e^{\lambda^2 t\|v\|^2}.\end{aligned}$$

For every sample path,

$$\psi_R(X_t) - \psi_R(X_0) \longrightarrow \lambda \langle v, X_t - X_0\rangle \qquad \text{as } R \to \infty.$$

The exponential is nonnegative. Fatou's lemma therefore gives

$$\mathbb{E}e^{\lambda\langle v, X_t - X_0\rangle} \le e^{\lambda^2 t\|v\|^2},$$

which proves theorem 1.1.

Remark 5.1 (Interpretation of the constant)**.** *The right-hand side is the moment-generating function of a centered Gaussian with variance $2t\|v\|^2$. Thus $X_t - X_0$ is sub-Gaussian with covariance proxy $2tI_d$. The conclusion does not assert that the increment itself is Gaussian. The constant reflects the normalization $\sqrt{2}\,dB_t$ in (1); with diffusion coefficient dB_t, the corresponding exponent would be divided by two.*

6 A concise Lyons–Zheng cross-check

For completeness, we record why the probabilistic literature gives the same estimate. Let $f_v(x) = \langle v, x\rangle$. The coordinate function belongs to the Dirichlet-form domain whenever it is square integrable under π, and

$$\mathcal{E}(f_v, f_v) = \|v\|^2.$$

For a symmetric process started from its symmetrizing measure, the Lyons–Zheng decomposition at the terminal time t gives

$$f_v(X_t) - f_v(X_0) = \frac{1}{2}\left(M_t^v - \widehat{M}_t^{v,t}\right),$$

where M^v is the forward martingale part and $\widehat{M}^{v,t}$ is the reverse-time martingale part. In the present Langevin diffusion,

$$[M^v]_t = [\widehat{M}^{v,t}]_t = 2t\|v\|^2,$$

and each terminal martingale is centered Gaussian because its integrand is the deterministic vector $\sqrt{2}v$ in its respective Brownian filtration. They are generally dependent. Cauchy–Schwarz, rather than independence, gives

$$\begin{aligned}\mathbb{E}e^{\lambda(f_v(X_t)-f_v(X_0))} &\le \left(\mathbb{E}e^{\lambda M_t^v}\right)^{1/2}\left(\mathbb{E}e^{-\lambda \widehat{M}_t^{v,t}}\right)^{1/2}\\ &= e^{\lambda^2 t\|v\|^2}.\end{aligned}$$

This reproduces theorem 1.1. The abstract theorem and the passage to unbounded coordinate functions are discussed in the references cited in Section 1.

7 Stopping times: what may and may not be substituted

The MALA appendix repeatedly writes a process at $t \wedge \tau$, where τ is a stopping time. This is legitimate, but not for the reason that an identity holds "for every deterministic t, almost surely" if the exceptional null set is allowed to depend on t. This section makes the distinction precise. Standard references for the stopping results used here are Karatzas and Shreve (1991, Chapter 3), Revuz and Yor (1999, Chapter IV), and Protter (2005, Chapter II).

7.1 Why fixed-time almost-sure equality is not enough

Suppose that, for every deterministic t, random variables Y_t and Z_t satisfy $Y_t = Z_t$ almost surely. This statement alone does not imply $Y_\tau = Z_\tau$ for a random time τ, because the exceptional null set may depend on t. A simple illustration is obtained on $\Omega = [0, 1]$ with Lebesgue measure: set

$$Y_t(\omega) = \mathbf{1}_{\{\omega=t\}}, \qquad Z_t(\omega) = 0, \qquad \tau(\omega) = \omega.$$

For each fixed t, $Y_t = Z_t$ almost surely, but $Y_\tau = 1$ everywhere. If one equips this space with the completed filtration generated by the events $\tau \leq s$ up to time t, then τ is a stopping time (and Y_t is adapted), so the pathology is not removed merely by the stopping-time property. This example is deliberately irregular, but it shows why the order of the quantifiers matters.

What is sufficient is an *indistinguishable process identity*: there is one event Ω_0 of probability one such that

$$Y_t(\omega) = Z_t(\omega) \quad \text{for every } t \geq 0 \text{ and every } \omega \in \Omega_0.$$

Then the equality may plainly be evaluated at $t = \tau(\omega)$. If two continuous processes agree almost surely at every rational time, continuity and countability imply that they are indistinguishable. This is why continuous versions are especially convenient.

7.2 Standard stopped-process identities

Let M be a continuous local martingale and τ a stopping time. The stopped process is

$$M_t^\tau = M_{t\wedge\tau}.$$

The following are standard process identities, valid up to indistinguishability:

$$[M^\tau]_t = [M]_{t\wedge\tau}, \tag{16}$$

and, if $M_t = \int_0^t \langle H_s, dB_s\rangle$,

$$M_{t\wedge\tau} = \int_0^t \mathbf{1}_{\{s\leq\tau\}} \langle H_s, dB_s\rangle . \tag{17}$$

The distinction between $s < \tau$ and $s \leq \tau$ is immaterial for a continuous Itô integral. Formula (17) can first be checked for simple predictable integrands and then extended by the Itô isometry. Formula (16) follows from the defining quadratic-variation theorem or from (17) when the martingale is a Brownian integral.

Ordinary algebraic identities between continuous processes may also be stopped. For example, if

$$D_t = \exp\{M_t - V_t/2\}$$

for every t on one probability-one event, then

$$D_{t\wedge\tau} = \exp\{M_{t\wedge\tau} - V_{t\wedge\tau}/2\}$$

on the same event. If V is increasing, then

$$V_{t\wedge\tau} \le V_t$$

pathwise. Neither statement uses optional sampling.

By contrast, an expectation identity such as $\mathbb{E}M_t = \mathbb{E}M_0$ for every deterministic t does not automatically imply $\mathbb{E}M_\tau = \mathbb{E}M_0$. That conclusion is an optional-sampling statement and requires its own hypotheses. The localized MALA proof uses the process identities above, not an unjustified optional-sampling step.

7.3 Line-by-line application to the path-likelihood lemma

For clarity, we match the stopping operations in the MALA proof to the results above. Let

$$M_t = \int_0^t \langle \theta_s, dB_s \rangle, \qquad V_t = [M]_t = \int_0^t \|\theta_s\|^2 \, ds, \qquad D_t = e^{M_t - V_t/2},$$

and let

$$\tau_n = \inf\{t : V_t \ge n\}.$$

(i) The identity

$$[M^{\tau_n}]_t = V_{t\wedge\tau_n}$$

is exactly (16). It is not obtained by taking a fixed-time equality and informally replacing t by a random time.

(ii) The factorization of $D^q_{h\wedge\tau_n}$ into a stopped stochastic exponential and an ordinary exponential is pointwise algebra applied to the indistinguishable process identity defining D. The inequality $V_{h\wedge\tau_n} \le V_h$ is pathwise because V is increasing.

(iii) Itô's formula gives the process identity

$$D_t - 1 = \int_0^t D_s \langle \theta_s, dB_s \rangle, \qquad t \ge 0,$$

up to indistinguishability. Stopping this identity at a further stopping time σ_n and using (17) gives

$$D_{t\wedge\sigma_n} - 1 = \int_0^t \mathbf{1}_{\{s\le\sigma_n\}} D_s \langle \theta_s, dB_s \rangle .$$

This is the square-integrable martingale to which BDG is applied.

(iv) The stopping is finally removed pathwise. On a fixed compact time interval, the continuous processes V and D are bounded on every sample path. Hence the stopping times eventually exceed the terminal horizon, and the stopped variables equal the unstopped variables for all large n. Fatou's lemma then transfers the uniform moment bound to the unstopped limit.

Thus the answer to the motivating question is: *partly, but the relevant assertion is stronger*. One needs process identities that hold simultaneously for all times outside one null set, or the standard theorem identifying the stopped process, stochastic integral, and quadratic variation. Merely knowing a separate almost-sure statement for each fixed t would not suffice.

8 Which parts are imported and which are specialized

The following table separates literature results from calculations specific to the MALA project.

Ingredient	Imported result	Specialization in the MALA paper
Stationary linear increments	Davies' exponential perturbation estimate; alternatively, the Lyons–Zheng decomposition	Apply the estimate to a truncated linear function and retain the exact constant for the $\sqrt{2}$-Langevin normalization.
Quadratic and integrated increments	Gaussian randomization is a standard device for turning linear sub-Gaussian bounds into quadratic-form bounds; compare Hsu et al. (2012, Theorem 1 and Remark 2)	Use an auxiliary standard Gaussian, then Jensen's inequality in time, to obtain the exact moment-generating and L^p bounds for $\int_0^h \|X_t - X_0\|^2\, dt$.
Path likelihood	Novikov, Girsanov, stopped stochastic exponentials, Doob, and quantitative BDG are standard. General reverse-Hölder theory for continuous stochastic exponentials is developed by Kazamaki (1994); related MALA diffusion comparisons appear in Chewi et al. (2021, Appendix A)	Choose the drift difference, prove the dimension- and moment-dependent step restriction, and exploit $V_h \le L^2 \int_0^h \|X_t - X_0\|^2\, dt/2$.
Endpoint conditioning	The likelihood ratio of a statistic is the conditional expectation of the full likelihood ratio; see, for example, Kallenberg (2021)	Condition the path likelihood on (X_0, X_h) and identify the exact and Euler endpoint measures.
Metropolis rejection	The accepted Metropolis–Hastings flow is the meet of proposal flow and its transpose (Tierney, 1998; Billera and Diaconis, 2001)	Use the symmetric exact endpoint measure as a dominating reference and bound the pointwise rejection function.

The first row is the only lemma in the MALA appendix that is essentially a direct specialization of one standard literature theorem. The remaining lemmas combine standard tools in a way adapted to the rejection problem; they should cite those tools and nearby diffusion-comparison literature, but should not be described as verbatim quotations.

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